

Week 6: Options Pricing & the Volatility Surface

Greeks, no-arbitrage, and SVI/SABR calibration

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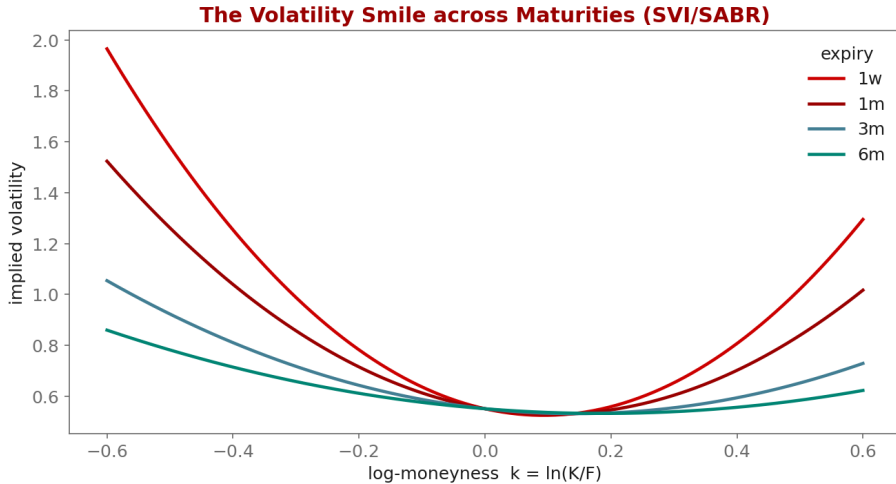
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1. Learning objectives
2. Lecture 1: Greeks & the Surface
3. Lecture 2: SVI & SABR Calibration
4. Illustration
5. Derivation & rationale
6. Getting the data from Coincall
7. Code: week6_bs_autograd.py
8. Summary & key takeaways
9. Glossary of key terms
10. Reading
11. References

- ▶ Recall the Greeks, including vanna and volga
- ▶ State the no-arbitrage constraints on the surface
- ▶ Calibrate SVI and SABR to Coincall data

- ▶ Black-Scholes and the Greek vector
- ▶ Vanna, volga, and volatility risk
- ▶ Calendar and butterfly arbitrage constraints

- ▶ Raw and arbitrage-free SVI
- ▶ SVI by nonlinear least squares (autograd Jacobians)
- ▶ The SABR / Hagan expansion



$$w(k) = a + b(\rho(k - m) + \sqrt{(k - m)^2 + \varsigma^2})$$

Why: Raw SVI for total implied variance $w = \sigma^2 T$ in log-moneyness k . The hyperbola shape is chosen because it matches observed smiles with only 5 parameters and has closed-form slopes for the no-arbitrage checks.

$$\text{butterfly: } g(k) \geq 0, \quad \text{calendar: } \partial_T w(k, T) \geq 0$$

Why: These inequalities are exactly 'no free option-portfolio arbitrage': a negative $g(k)$ implies a negative risk-neutral density, and decreasing total variance in T implies a calendar arbitrage. We enforce them so the fitted surface is tradeable.

- ▶ Pull a full option chain with mark IV and Greeks per strike:
 - ▶ Chain: `GET /open/option/get/v1/{index}` (index = BTC/ETH underlying)
 - ▶ Detail: `GET /open/option/detail/v1/{symbol}` (mark, IV, delta/gamma/vega/theta/rho)
 - ▶ Symbols: `GET /open/option/getInstruments/BTC`
- ▶ Offline: `coincall_btc_options_chain_snapshots_2025Q4.parquet` (5-min snapshots)
- ▶ A baseline `coincall_calibrated_svi_surfaces_2025Q4.parquet` is provided

Cross-check exact paths at <https://docs.coincall.com/>

```

import torch
def bs_call(S, K, T, r, sig):
    d1 = (torch.log(S/K) + (r + 0.5*sig**2)*T) / (sig*torch.sqrt(T))
    d2 = d1 - sig*torch.sqrt(T)
    N = torch.distributions.Normal(0., 1.).cdf
    return S*N(d1) - K*torch.exp(-r*T)*N(d2)

S,K,T,r,sig = [torch.tensor(x, requires_grad=True)
               for x in (100., 100., 1.0, 0.0, 0.6)] # crypto IV ~ 60%
price = bs_call(S,K,T,r,sig)
vega, = torch.autograd.grad(price, sig, create_graph=True)
vanna, = torch.autograd.grad(vega, S, create_graph=True) # d vega / dS
volga, = torch.autograd.grad(vega, sig) # d vega / d sig
print(float(price), float(vega), float(vanna), float(volga))

```

- ▶ Implied volatility forms a smile; SVI and SABR parameterize it
- ▶ Fit with PyTorch autograd Jacobians – no hand-derived gradients
- ▶ Enforce calendar and butterfly no-arbitrage so the surface is tradeable
- ▶ Greeks (incl. vanna, volga) come straight from autograd
- ▶ Crypto data is noisy: weight by volume/spread and seed from yesterday's fit

Implied volatility the volatility that matches the market option price under Black-Scholes

Volatility smile implied volatility as a function of strike / moneyness

SVI Stochastic Volatility Inspired parameterization of total variance

SABR stochastic alpha-beta-rho model with the Hagan implied-vol expansion

Greeks price sensitivities: delta, gamma, vega, theta, rho

Vanna / Volga second-order Greeks: $d \text{ vega} / dS$ and $d \text{ vega} / d \sigma$

Calendar / butterfly arbitrage no-arbitrage constraints the surface must satisfy

- ▶ Gatheral (2006), Ch. 1-5; Gatheral-Jacquier (2014); Hagan et al. (2002)

- ▶ Gatheral, J. (2006). *The Volatility Surface: A Practitioner's Guide*. Wiley.
- ▶ Gatheral, J., & Jacquier, A. (2014). Arbitrage-free SVI volatility surfaces. *Quantitative Finance*, 14(1), 59-71.
- ▶ Hagan, P. S., Kumar, D., Lesniewski, A. S., & Woodward, D. E. (2002). Managing smile risk. *Wilmott Magazine*, Sept., 84-108.
- ▶ Hull, J. C. (2021). *Options, Futures, and Other Derivatives* (11th ed.). Pearson.